3.3 Changes of state

Change of state of a substance is one of the major physical changes we encounter in daily lives. We have read about simple changes of physical state such as melting of ice in our previous classes.

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The following are some of the changes of state:

from Solid → to Liquid is Melting

from Liquid → to Gas is Vaporization

from Liquid → to Solid is Freezing

from Gas → to Liquid is Condensation

from Solid → to Gas is Sublimation

Melting, vaporization, and sublimation occur when heated and hence it is called as endothermic process. In an endothermic process, the speed of the molecules is increased hence they move faster.

In contrast, such as in freezing and condensation, heat is removed, resulting in the decreasing the speed of the molecules causing them move slower. Such processes are called as exothermic process.

In the next section we will look at each of these physical changes.

3.3.1 Melting

You have seen a puddle of water getting pooled around the glass of ice-cream or a glass of ice cubes when it is kept in room temperature. The ice cubes / ice-cream melt. Right! Can you give reason for that? The ice kept in the beaker receives heat from the surrounding air, to melt and form water.

ACTIVITY 1

Melting of ice and freezing of water

Though ice and water look different, they are both made of water molecules. This means that no new substance is formed during the melting of ice, only a change of state from solid to liquid takes place during the melting of ice. So, the melting of ice to form water is a physical change.

The change which occurs during the melting of ice to form water can be reversed easily by freezing the water to form ice again by keeping a beaker of water in the freezer zone of a refrigerator.

Thus we can find that

$$\text{Solid} \xrightarrow{\text{heating}} \text{Liquid}$$

$$\text{Liquid} \xrightarrow{\text{cooling}} \text{Solid}$$

Melting is the changing of a solid into its liquid state and it happens by heating, whereas Freezing is the changing of a liquid into its solid state and it happens by cooling.

3.3.2 Vapourization

Look at a kettle kept on the fire. The bubbles form and the liquid water becomes

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